

Cambrian-*P*: Pose-Grounded Video Understanding

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用Camera Pose Supervision提升Video MLLM空间智能

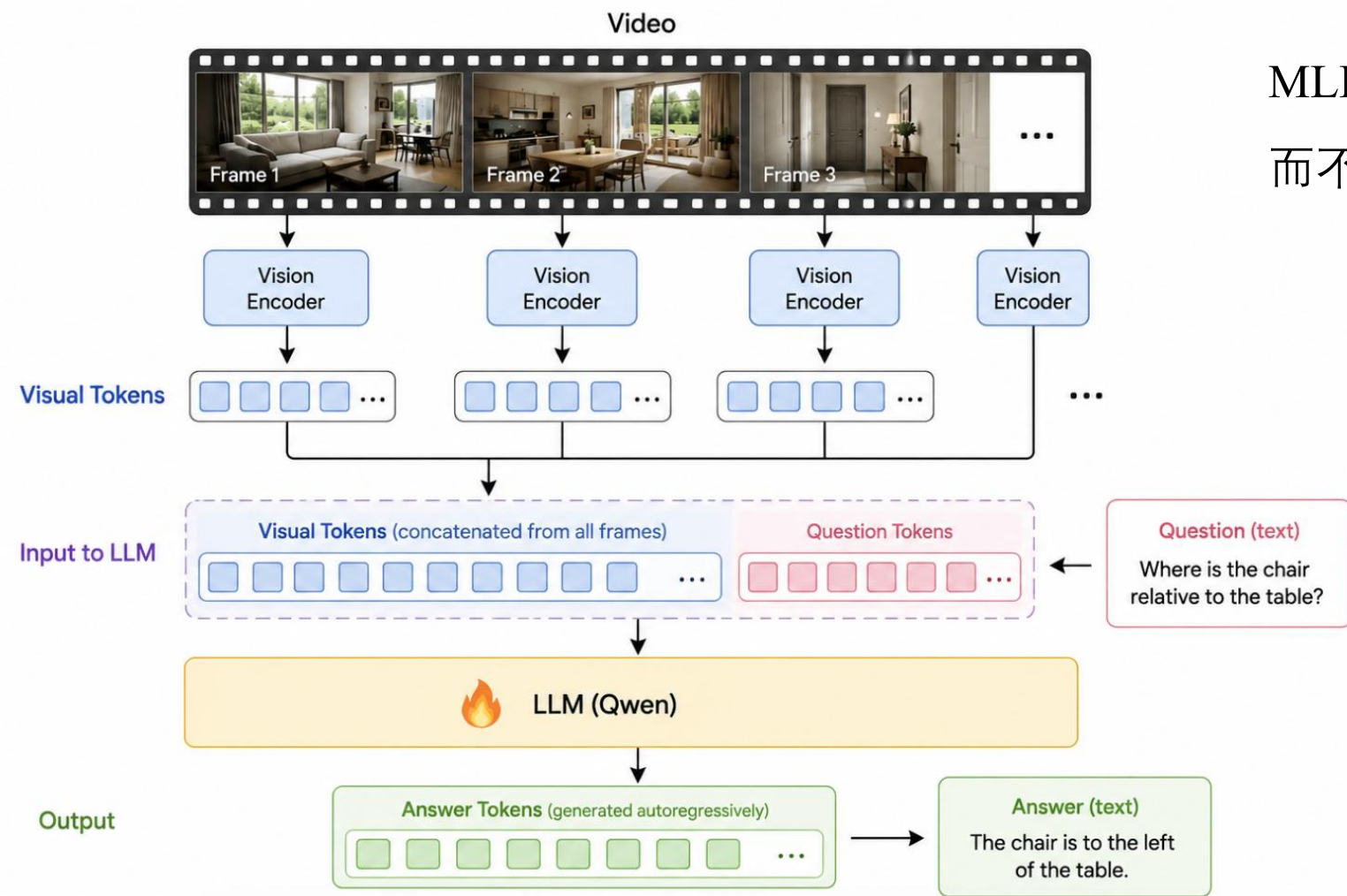
<https://cambrian-mlm.github.io/cambrian-p/>

王锦

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Task&Background

Video MLLM缺少空间理解能力



MLLM看到的是孤立的帧

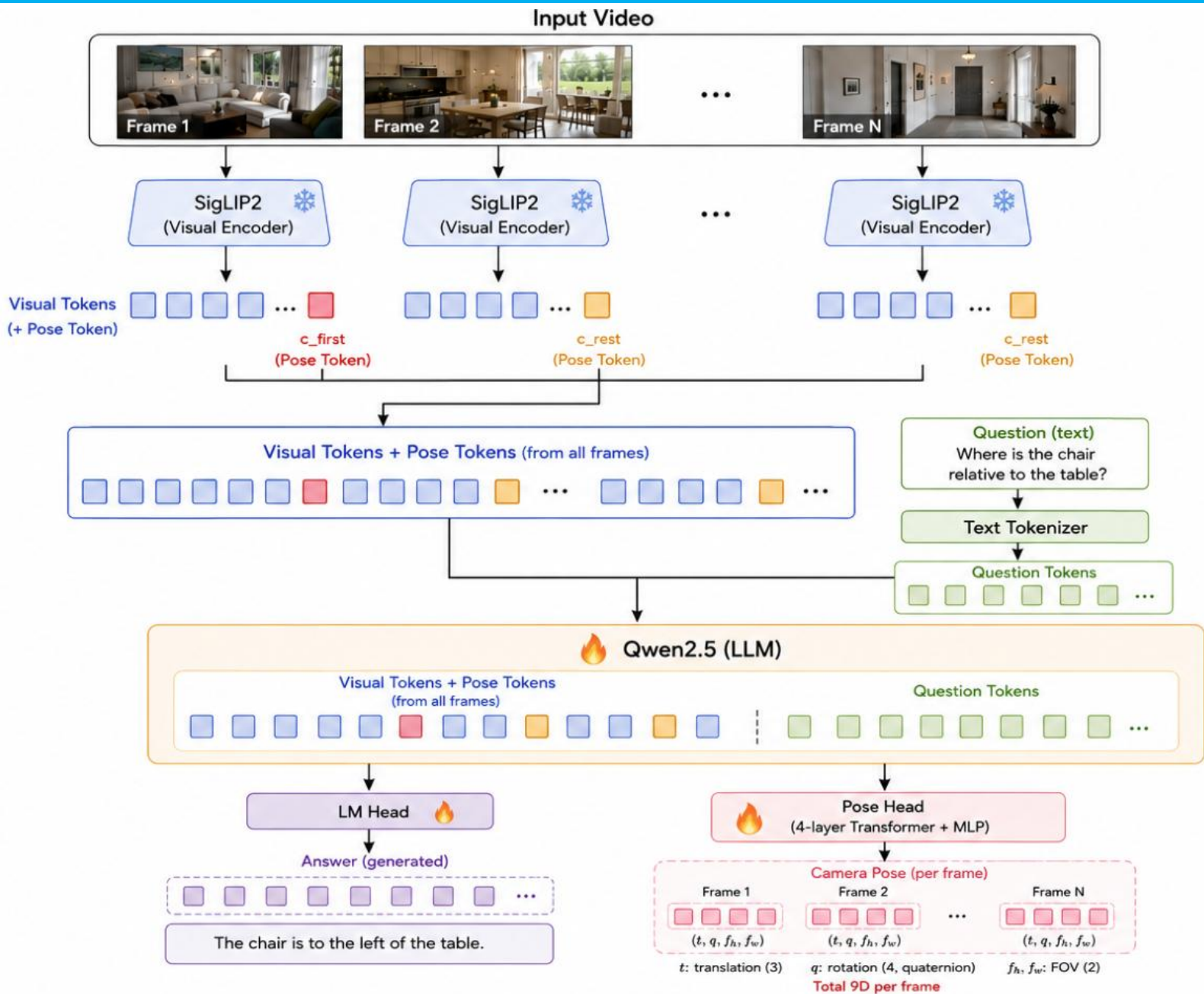
而不知道Frame1 和 Frame2在空间上如何对应

失败案例

- Relative Direction
- Route Planning
- Distance Estimation

Video MLLM 缺少统一3D坐标系，无法建立跨帧空间对应关系。

Overall Framework



Cambrian-S:

SigLIP2 + MLP Projector + Qwen2.5

Cambrian-P只新增了:

1.Pose Token

2.Pose Head

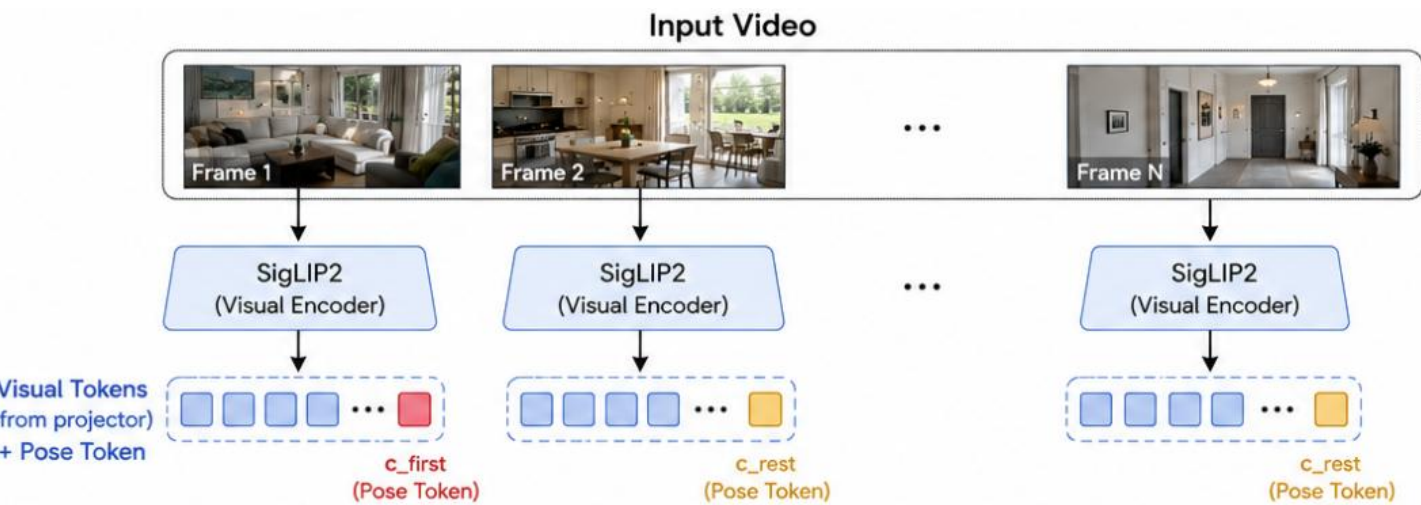
没有额外3D Encoder。

Architecture I: Pose Token

定义： c_first c_rest两个可学习向量。

c_first: 第一帧=参考坐标系

c_rest: 其余帧=相对于第一帧的偏移

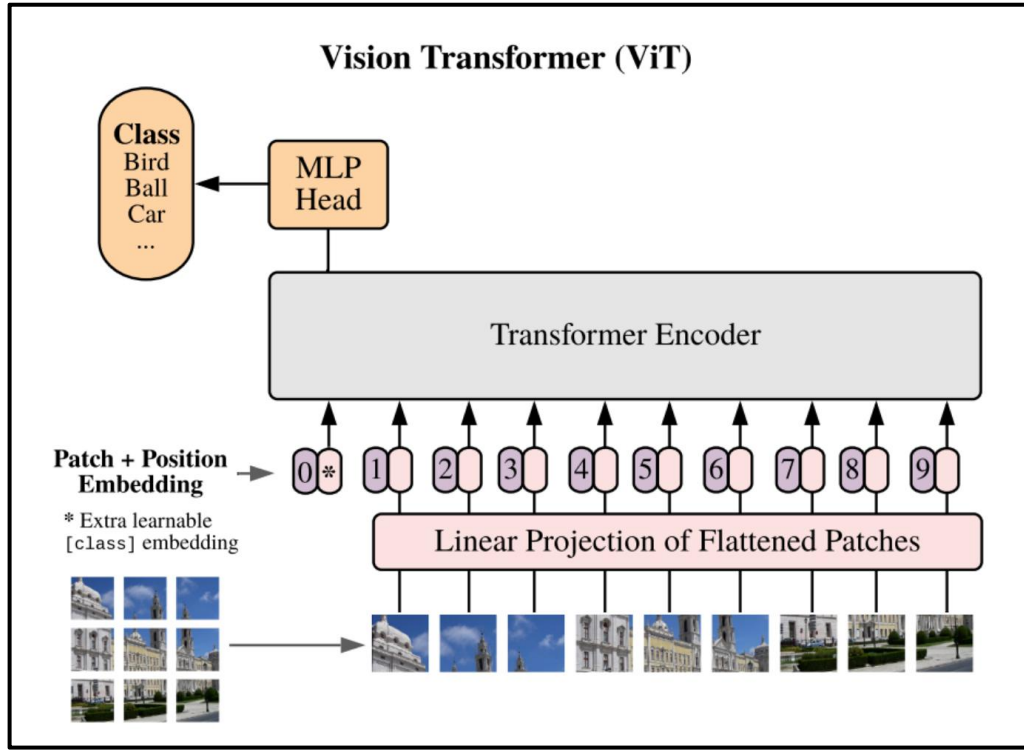


Pose Token为什么放在Visual Token后面？

mask multi-head self-attention

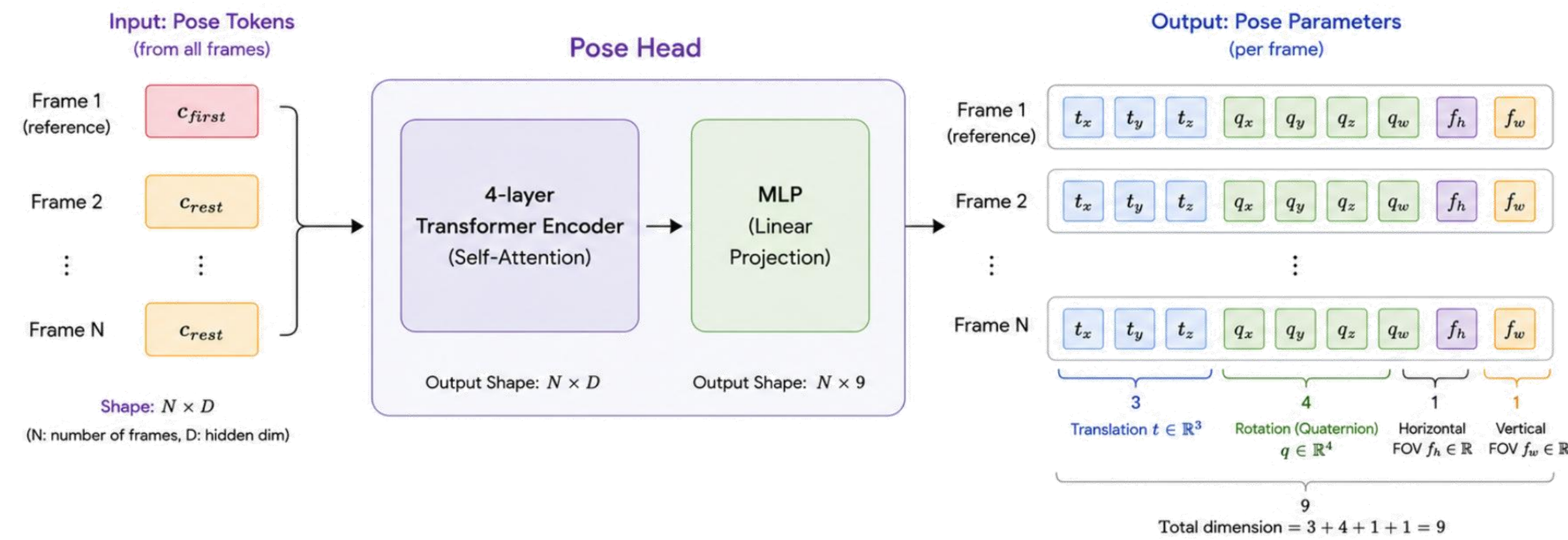
Pose Token可以聚合整帧视觉信息。

类似CLS Token， 变成这一帧的压缩表示。



Architecture II: Pose Head

经过位姿回归损失监督后，Pose Tokens就编码了相机朝向、移动方向、场景布局、空间结构等3D几何感知特征



Data

数据集	Video	QA	Pose	Usage
VSI-590K（Pose子集）	✓	✓	✓	Joint Supervision
VSI-590K（非Pose子集）	✓	✓	✗	VQA-only
MapAnything	✓	✗	✓	Pose-only

Video QA 和 Camera Pose Estimation的最优数据采样、数据增强策略以及训练时长是冲突的：

1. 帧采样冲突

QA 通常“均匀采样”； Pose 通常“随机起点 + 动态间隔”。

2. 数据增强冲突

QA 希望图像尽量“真实不变”； Pose 反而依赖强数据增强提升鲁棒性。

3. 训练轮次

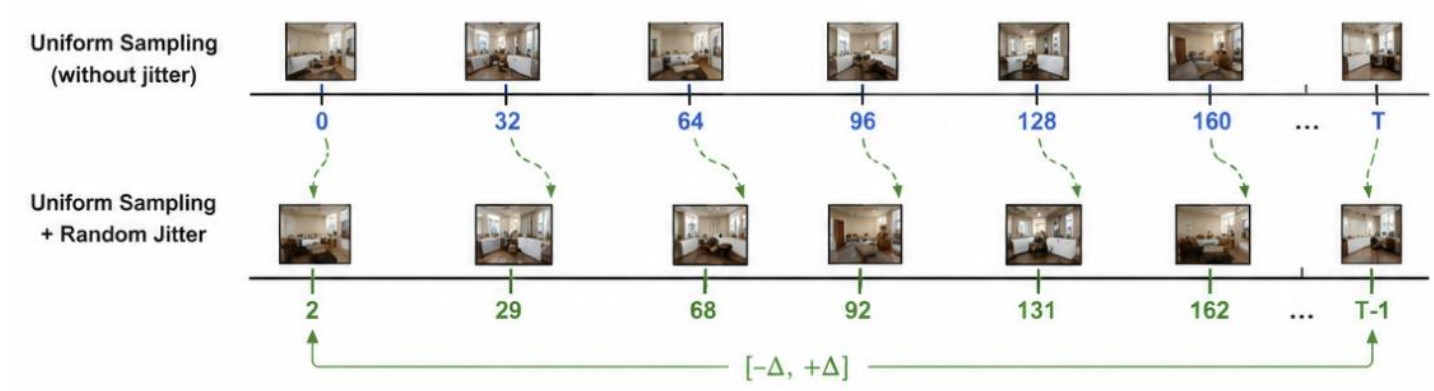
MLLMs通常仅训练 1 个周期， 而姿态估计模型则需数十个周期才能收敛。

Training

Interleaved Training (交替训练)



Random Jitter (随机抖动)



$$L = L_{NTP} + \lambda_{pose} L_{pose}$$
$$L_{pose} = L_T + L_R + L_{FOV}$$

Type	Sampling	Augmentation	Loss
VQA-only	Uniform Sampling	None	L_{NTP}
Joint Supervision	Random Jitter	Weak	$L_{NTP} + \lambda_{pose} L_{pose}$
Pose-only	Random Start + Dynamic Interval	Strong	L_{pose}

让 QA batch 和 Pose-only batch 各自采用最优的数据采样与增强策略

Results: Spatial Understanding

VSI-Bench

Model	LM	Avg.	Numerical Answer				Multiple-Choice Answer			
			Obj. Count	Abs. Dist.	Obj. Size	Room Size	Rel. Dist.	Rel. Dir.	Route Plan	Appr. Order
<i>Baselines</i>										
Chance Level (Random)	–	–	–	–	–	–	25.0	36.1	28.3	25.0
Chance Level (Frequency)	–	34.0	62.1	32.0	29.9	33.1	25.1	47.9	28.4	25.2
<i>General-purpose Models</i>										
GPT-4o [41]	Unk.	34.0	46.2	5.3	43.8	38.2	37.0	41.3	31.5	28.5
Gemini-2.5 Pro [25]	Unk.	51.5	43.8	34.9	64.3	42.8	61.1	47.8	45.9	71.3
Qwen2.5VL-7B [8]	Qwen2.5-7B	29.3	25.2	10.5	36.4	29.6	38.4	38.0	29.8	26.8
InternVL-3 8B [129]	Qwen2.5-7B	42.1	68.1	39.0	48.4	33.6	48.3	36.4	27.3	35.4
InternVL-3.5 8B [129]	Qwen3-8B	56.3	–	–	–	–	–	–	–	–
Qwen3-VL 8B [7]	Qwen3-8B	56.6	–	–	–	–	–	–	–	–
<i>Spatial-specialist Models</i>										
VST 7B [110]	Qwen2.5-7B	61.2	71.6	43.8	75.5	69.2	60.0	55.6	44.3	69.2
VLM-3R 7B [29]	Qwen2-7B	60.9	70.2	49.4	69.2	67.1	65.4	80.5	45.4	40.1
VG-LLM 8B [128]	Qwen2.5-7B	50.7	67.9	37.7	58.6	62.0	46.6	40.7	32.4	59.2
Cambrian-S 7B [111]	Qwen2.5-7B	67.5	73.2	50.5	74.9	72.2	71.1	76.2	41.8	80.1
SenseNova-SI 8B [17]	Qwen2.5-7B	68.7	–	–	–	–	–	–	–	–
GeoThinker 7B [48]	Qwen2.5-7B	68.5	+4.5	–	–	–	–	–	–	–
GeoThinker 8B [48]	Qwen3-8B	72.6		–	–	–	–	–	–	–
Cambrian-S-7B [†] [111]	Qwen2.5-7B	69.2	73.6	53.7	75.2	74.7	71.5	82.0	38.7	84.3
Cambrian-P	Qwen2.5-7B	73.7	74.9	60.1	76.0	76.9	74.8	89.5	52.6	85.0

VSTemporalI-Bench

Methods	Avg.	Cam-Obj Abs. Dist.	Cam. Displace.	Cam. Mov. Dir.	Obj-Obj Rel. Pos.	Cam-Obj Rel. Dist.
GPT-4o [41]	38.2	29.5	23.4	37.3	58.1	42.5
Gemini-1.5 Flash [87]	32.1	28.5	20.9	24.4	52.6	33.9
LLaVA-NeXT-Video-72B [57]	44.0	32.3	10.5	48.1	78.3	50.9
VLM-3R-7B [29]	58.8	39.4	39.6	60.6	86.5	68.6
GeoThinker 8B [48]	67.4	38.4	45.8	84.2	93.6	75.2
Cambrian-P (w/o Pose)	62.4	39.4	40.6	67.7	92.2	72.0
Cambrian-P	68.9	42.5	46.6	87.7	94.3	73.2

显著增强相机运动理解能力

Results: Camera pose estimation

Cambrian-P达到接近专业3D系统的水平

Model	ScanNet			TUM-dynamic			Sintel		
	ATE ↓	RPE trans ↓	RPE rot ↓	ATE ↓	RPE trans ↓	RPE rot ↓	ATE ↓	RPE trans ↓	RPE rot ↓
Offline Models									
VGGT [96]	0.035	0.015	0.380	0.009	0.008	0.350	0.172	0.061	0.470
DUST3R-GA [99]	0.081	0.028	0.784	0.083	0.017	3.567	0.417	0.250	5.796
MASt3R-GA [66]	0.078	0.020	0.475	0.038	0.012	0.448	0.185	0.060	1.496
MonST3R-GA [122]	0.077	0.018	0.529	0.098	0.019	0.935	0.111	0.044	0.869
Fast3R [107]	0.155	0.123	3.491	0.090	0.101	1.425	0.371	0.298	13.750
FLARE [124]	0.064	0.023	0.971	0.026	0.013	0.475	0.207	0.090	3.015
π^3 [102]	0.031	0.013	0.347	0.014	0.009	0.312	0.074	0.040	0.282
MapAnything [44]	0.052	0.025	0.720	0.029	0.023	0.370	0.226	0.077	0.640
Streaming Models									
StreamVGGT [130]	0.127	0.041	1.880	0.062	0.030	0.690	0.273	0.109	0.850
CUT3R [98]	0.096	0.022	0.590	0.045	0.015	0.440	0.215	0.070	0.630
Point3R [103]	0.097	0.035	2.791	0.058	0.031	0.758	0.442	0.154	1.897
Spann3R [95]	0.096	0.023	0.661	0.056	0.021	0.591	0.329	0.110	4.471
G ² VLM [37]	0.148	0.048	1.220	0.129	0.044	0.700	0.301	0.135	1.450
Cambrian-P	0.078	0.023	0.880	0.046	0.020	0.580	0.239	0.081	2.440

Cambrian-P参数量显著更多，但在离线和流式两种场景下均实现最低延迟

Method	#Params	Offline		Streaming	
		Per-sequence (s)	Per-frame (s) [†]	Per-sequence (s)	Per-frame (s)
VGGT [96]	1.26B	9.90	0.11	—	—
CUT3R [98]	0.80B	5.22	0.06	6.03	0.07
StreamVGGT [130]	1.26B	—	—	9.00	0.10
Cambrian-P (Ours)	8.20B	2.16	0.02	5.76	0.06

Ablation

Table 8 | **Components ablation study of Cambrian-*P*.**

ATE: 相机姿态估计的平均绝对误差

Camera Loss	Interleaved Training	Random Jitter	VSI-Bench ↑	ScanNet ATE↓	TUM ATE ↓	Sintel ATE ↓
			67.3	-	-	-
✓	✓		71.2	0.144	0.106	0.366
✓		✓	69.4	0.259	0.132	0.521
✓	✓	✓	72.0	0.141	0.096	0.325

Pose Loss	Depth Loss	VSI-Bench ↑	ScanNet ATE↓	TUM ATE ↓	Sintel ATE ↓
✗	✗	67.3	-	-	-
✓	✗	72.0	0.141	0.096	0.325
✓	✓	71.7	0.156	0.118	0.318
✗	✓	69.4	0.371	0.194	0.537
T only	✗	70.7	0.205	0.128	0.357
R only	✗	69.7	0.287	0.092	0.353
FV only	✗	69.4	0.408	0.196	0.670
T + R	✗	71.5	0.165	0.130	0.386

Camera Pose 是一种比 Depth 更适合
Video MLLM 的 3D Supervision Signal

Ablation

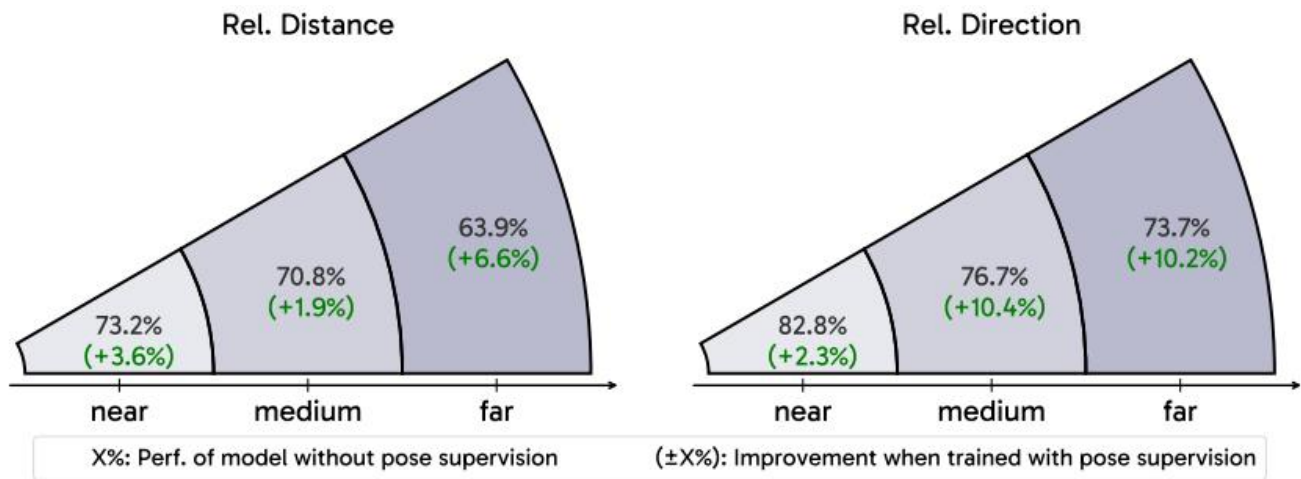
推理时不需要Pose

Pose Training	Token Inference	VSI-Bench	VSTIBench	MVBench	EgoSchema	Perception Test	Tomato
✗	✗	67.3	55.1	51.9	49.6	56.4	20.4
✓	✓	72.0	56.5	53.5	52.5	58.4	26.7
✓	✗	72.0	56.6	53.2	52.5	58.8	26.7

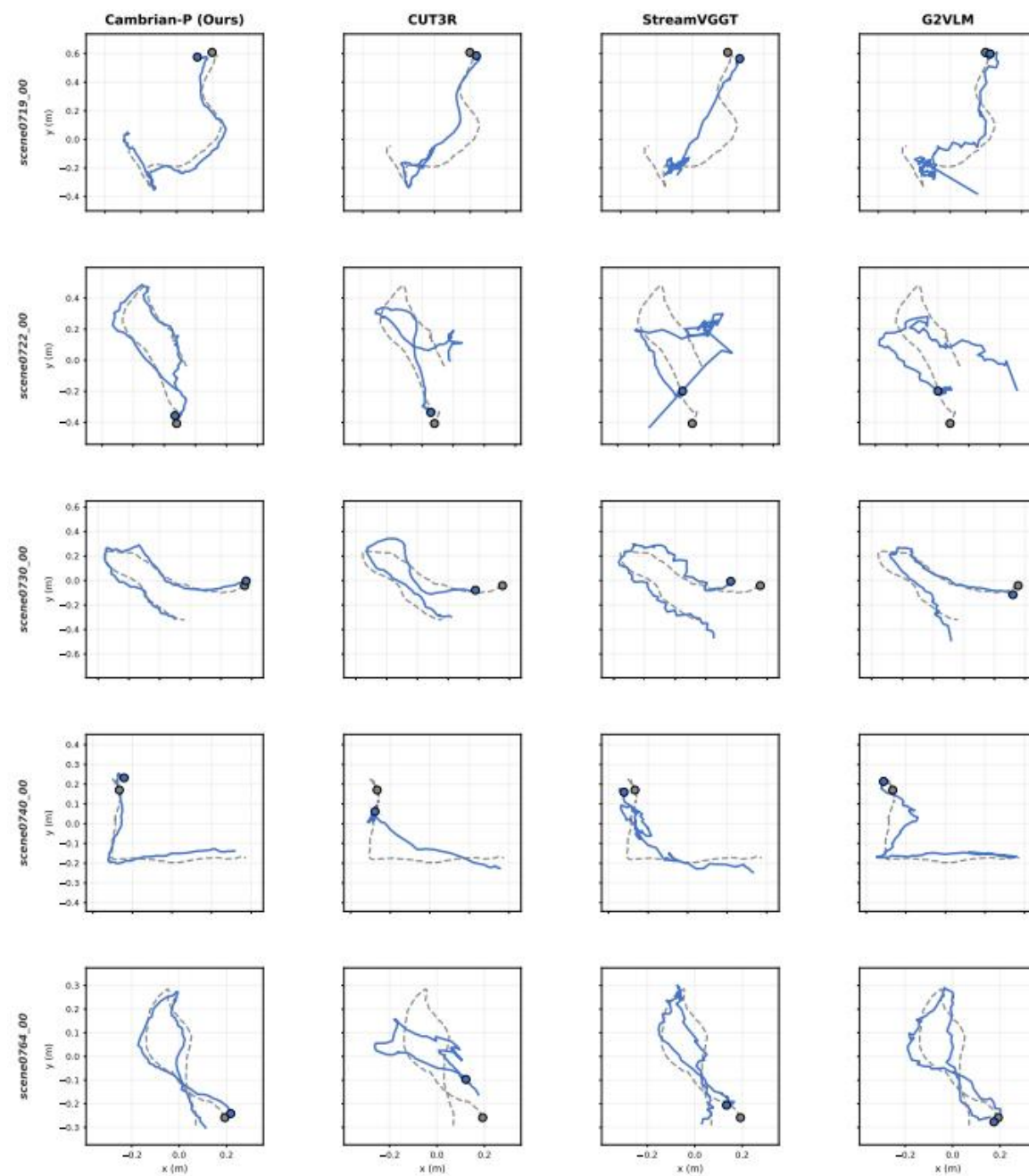
Pose 是一种 supervision signal，而非Inference Signal

- 1. 提高推理效率
- 2. Better Representation Learning

Analysis&Visualization



Cambrian-P模型对远处物体的推理效果显著优于附近物体。
表明Pose监督能帮助MLLMs提升全局空间推理能力。



Conclusion

- Pose Token = Video版CLS Token, 有效3D监督信号
- 对于 Video MLLM $\text{Pose} > \text{Depth}$
- 训练学Pose 推理不需要Pose
- Interleaved Training

